500. Let the number of cats be x. Then, number of dogs = $55 - x$.

Clearly, x is a multiple of 5 i.e., $x = 5n$ where $n \le 10$.

And, $(55 - x)$ is a multiple of 4 $\Rightarrow 55 - 5n = 4m$

$$\Rightarrow m = \frac{55 - 5n}{4}.$$

Since m is a natural number, $(55 - 5n)$ must be a multiple of 4 which happens for $n = 3$ or 7.

Case I. If $n = 3$, $m = 10$.

Number of cats = 15, Number of dogs = 40.

Number of pets adopted = $\left(\frac{1}{5} \times 15 + \frac{1}{4} \times 40\right) = 13.$

Case II. If $n = 7$, $m = 5$.

Number of cats = 35, Number of dogs = 20.

Number of pets adopted = $\left(\frac{1}{5} \times 35 + \frac{1}{4} \times 20\right) = 12.$

$\therefore$ Required number = 13.

501. The team has played a total of $(17 + 3) = 20$ matches. This constitutes $\frac{2}{3}$ of the matches. Hence, total number of matches played = 30. To win $\frac{3}{4}$ of them, a team has to win at least 23 of them. The team has thus to win a minimum of 6 matches out of remaining 10. So, it can lose a maximum of 4 of them.

502. Let M represent the number of mornings when they swam.

Then, total number of mornings in the vacation

$= 32 + M$.

Let E represent the number of evenings when they played tennis.

Then, total number of evenings in the vacation = $18 + E$.

$\therefore \quad 32 + M = 18 + E \Rightarrow E - M = 14.$ ...(*i*)

Total number of days when they swam or played

$= E + M$.

$\therefore \quad E + M = 28$ ...(*ii*)

Adding (*i*) and (*ii*), we get: $2E = 42$ or $E = 21$.

Hence, duration of vacations = $18 + E = 18 + 21 = 39$.

503. **Note:** $10 \times 15 \neq 20 \times 10$. So, this is not a question of Chain Rule and the original quantity of grass and its growth rate are to be considered.

Let g be the total units of grass in the field originally, r be the units of grass grown per day and c be the units of grass each cow eats per day.

Then, $g + 15r = (10 \times 15)\, c = 150c$ ...(*i*)

$g + 10r = (20 \times 10)\, c = 200c$ ...(*ii*)

Subtracting (*ii*) from (*i*), we get: $5r = -50c$ or $r = -10c$.

Putting $r = -10c$ in (*i*), we get: $g = 300c$.

Let the required number of days be x.

Then, $g + xr = 30xc \Rightarrow 300c - 10xc = 30xc \Rightarrow 40xc = 300c$

$$\Rightarrow 40x = 300 \Rightarrow x = 7\frac{1}{2}.$$

504. Let g be the total units of grass in the field originally, r be the units of grass grown per day and c be the units of grass each cow eats per day.

Then, units of grass consumed by 70 cows in 24 days

$= (70 \times 24 \times c) = 1680c$.

Units of grass consumed by 30 cows in 60 days

$= (30 \times 60 \times c) = 1800c$.

We have: $g + 24r = 1680c$ ...(*i*)

$g + 60r = 1800c$ ...(*ii*)

Subtracting (*i*) from (*ii*), we get: $36r = 120c \Rightarrow r = \frac{10}{3}c.$

Putting $r = \frac{10}{3}c$ in (*i*), we get: $g = 1600c$.

Let the required number of cows be x.

Then, $g + 96r = x \times 96 \times c \Rightarrow 1600c + \frac{96 \times 10}{3}c = 96xc$

$\Rightarrow 1920c = 96xc \Rightarrow 96x = 1920 \Rightarrow x = 20.$

Questions 505 and 506

Let the total number of pilgrims in the group be x.

Then, total number of visits to shrines

= Number of shrines × Number of pilgrims visiting each shrine = $10 \times 4 = 40$.

So, $5x = 40$ or $x = 8$.

Let the number of lakes be n.

Then, total number of visits to lakes

= Number of lakes × Number of pilgrims visiting each lake = $6n$.

$\therefore \quad 6n = 8 \times 3 = 24$ or $n = 4$.

505. Number of lakes in the city = 4.

506. Number of pilgrims in the group = 8.

507. Let the total number of apples be x. Then,

Apples sold to 1st customer = $\left(\frac{x}{2} + 1\right)$. Remaining apples

$$= x - \left(\frac{x}{2} + 1\right) = \left(\frac{x}{2} - 1\right).$$

Apples sold to 2nd customer = $\frac{1}{3}\left(\frac{x}{2} - 1\right) + 1$

$$= \frac{x}{6} - \frac{1}{3} + 1 = \left(\frac{x}{6} + \frac{2}{3}\right).$$

Remaining apples = $\left(\frac{x}{2} - 1\right) - \left(\frac{x}{6} + \frac{2}{3}\right)$

$$= \left(\frac{x}{2} - \frac{x}{6}\right) - \left(1 + \frac{2}{3}\right) = \left(\frac{x}{3} - \frac{5}{3}\right).$$

Apples sold to 3rd customer = $\frac{1}{5}\left(\frac{x}{3} - \frac{5}{3}\right) + 1 = \left(\frac{x}{15} + \frac{2}{3}\right).$

Remaining apples = $\left(\frac{x}{3} - \frac{5}{3}\right) - \left(\frac{x}{15} + \frac{2}{3}\right)$

$$= \left(\frac{x}{3} - \frac{x}{15}\right) - \left(\frac{5}{3} + \frac{2}{3}\right) = \left(\frac{4x}{15} - \frac{7}{3}\right).$$

$$\therefore \quad \frac{4x}{15} - \frac{7}{3} = 3 \Leftrightarrow \frac{4x}{15} = \frac{16}{3} \Leftrightarrow x = \left(\frac{16}{3} \times \frac{15}{4}\right) = 20.$$

508. Let the time taken to read one page of Engineering Mathematics be M units and that taken to read one page of Engineering Drawing be D units.

Then, $80\,M + 100\,D = 50\,M + 250\,D \Rightarrow 30\,M = 150\,D \Rightarrow D = \frac{M}{5}.$